

Energy Transformations in Column of Plasma-Focus Discharges With Megaampere Currents

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Abstract—Energy transformations are estimated from experimental results obtained on the PF-1000 plasma focus facility operated with a deuterium gas filling and maximum current of 2 MA, which enabled the neutron yield of $> 10^{11}$ to be obtained. Use was made of the laser interferometry images and the recorded signals of hard X-rays (HXR) and neutrons, as well as voltage, current, and current-derivative waveforms. The calculations of the inductance, derivative of inductance, and resistance of the current sheath enabled the evaluation of energy delivered to the plasma. It was estimated that, during the intensive pulses of the HXR and neutrons, energy of 3–11 kJ is delivered to the plasma column probably through an anomalous resistance at the time closely before and during pulses of HXR and neutron production. A predominant portion of this energy is effectively used for the acceleration of fast deuterons capable of producing fusion neutrons.

Index Terms—Anomalous resistance, interferometric diagnostics, neutron sources, plasma focus, plasma pinch.

I. INTRODUCTION

ONE OF the important reasons to study Z-pinch discharges (including PF-type ones) is the efficient generation of fast electrons and ions of energies ranging from tens of kiloelectronvolts to hundreds of kiloelectronvolts. At the deuterium filling, the investigation of fast ions can be indirectly provided by fast neutrons and protons produced from the D–D fusion reactions [1]–[6]. Various mechanisms of the acceleration of charged particles to high energy are still under discussion. A list of the suggested hypotheses was reviewed in other papers, e.g., [7] and [8].

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The X-ray and neutron studies within the PF-1000 plasma-focus facility at the Institute of Plasma Physics and Laser Microfusion in Warsaw, Poland, were performed at the current level of 1–2 MA and a neutron yield of above 10^{11} . Some recent results have already been reported in papers [5] and [10]–[12]. The plasma was investigated with a temporally and spatially resolved X-ray and neutron diagnostics. Signals obtained from scintillation detectors enabled estimations of a period of the neutron production and neutron energy distribution using the time-of-flight method [8]. From the spatial distribution of neutron energies, the energy of fast deuterons producing neutrons was evaluated [11]. These deuterons had the dominant energy component of 70–150 keV in the downstream direction (i.e., along the motion of the current sheath) and a considerable component (of 40–100 keV) in the radial direction. Then, the neutrons are produced by collisions of fast deuterons partially circulating in the column with the thermal deuterons.

In later papers [13]–[15], the interferometric diagnostics were used for an analysis of the evolution of the plasma density and geometry of the dense column, with the temporal resolution of 10–20 ns and the spatial resolution of about 1 mm. The hard X-rays (HXR) and neutrons were produced during four different periods: 1) during the formation and decay of a dense spherical structure (plasmoid) at the initial phase of the plasma stagnation; 2) during a decrease in the plasma density in the region between the anode and the plasmoid (in the stagnation phase); 3) during evolution of a constriction (phase of development of instabilities); and 4) during the integration of a “plasma lobule” with the pinch column (the “plasma lobule” means a local 2-D or 3-D deformation which prolapses the surface of the dense plasma column). The neutron sources can be identified with dense plasmoids. They correspond to the spherical source of fusion protons registered with a pinhole camera [12] as to the densest region of the plasma column registered in the time of neutron production, in which the highest probability of D–D fusion reaction occurs [13], [14]. The acceleration of fast deuterons, observed to be mainly directed along the motion of the current sheath, must be taking place in the region between the anode and the plasmoid. In this region, a fast decrease of plasma density is observed [13], [14]. In [15], the total number (about 10^{17}) of fast deuterons producing observed neutrons in intense pulses was estimated to have total energy of 3–5 kJ. It was suspected that the plasma evolution is connected with a spontaneous transformation of azimuthal and self-generated poloidal components of magnetic fields by reconnections [15].

This paper deals with transformation of the energy from the condenser battery into the plasma. The method of this transport is unclear at this time. There are two possibilities for the increase of the internal energy: increase of the inductance and/or of the resistance. Inductance in the plasma column increases both with the decrease of the internal radius of the plasma-carrying current as well as the more complicated direction of this current in the plasma column. The second possibility occurs at the increase of the resistance. The temporally resolved interferometry and neutron diagnostics enabled an evaluation of the energy transformation at the time of deuteron acceleration. For discussion of the energy balance (Section II-A), it is necessary to know the waveforms of current, voltage, current derivative, inductance, and electron temperature (for evaluation of Spitzer resistance). The characteristics of the first three values are presented in Section II-B. The inductance can be calculated from the known geometry of electrodes and the profile of the plasma column imaged by interferometry with the assumption of a thin skin layer and current flowing at the surface layer of the dense plasma column (Section II-C). In Section II-C, we evaluate temperature, radiation, and resistance. Furthermore, we evaluate the value of energy coming into plasma, the kinetic energy of the imploding plasma sheath, and the energy of a magnetic field in the dense plasma column. We compare the time and value of the energy input with the time and energy of the escaping fast deuterons capable of producing fusion neutrons in Sections II-D–F. The results are discussed and summarized in Section III.

II. EXPERIMENTAL RESULTS AND DISCUSSION

The described measurements were performed within the PF-1000 facility equipped with the Mather-type coaxial electrodes of 480 mm in length. The copper anode was 230 mm in diameter with a central hole of 35-mm diameter. The cathode of 400 mm in diameter was composed of 12 stainless-steel tubes, each of 80-mm diameter. The facility was operated at the charging voltage equal to 27 kV, stored energy equal to about 470 kJ, current maximum of 1.8–2.0 MA, and current during the pinch phase amounting to 1.2–1.4 MA. The initial pressure of the deuterium filling was 250–300 Pa.

The voltage, current, and current-derivative signals were measured at the current collector, near the main insulator. X-ray pulses were recorded with a silicon PIN detector shielded with a Be foil of 10 μm in thickness. Four scintillation detectors coupled with fast photomultipliers were used to perform detailed time-resolved measurements of HXR and neutrons. Three probes were situated at a distance of 7.0 m downstream, upstream, and side-on. The fourth probe was placed at a distance of 16 m in the upstream direction. The interferometric measurements were performed with a Nd:YLF laser working at the second harmonics (527 nm). One laser pulse (lasting below 1 ns) was split by a setup of mirrors into 16 separated beams, which penetrated through a Mach–Zehnder interferometer and could be used to investigate the plasma region in the interval 0–220 ns with interframe delays of 10–20 ns. From the recorded interferometric images, the distribution of the electron density along the direction perpendicular to z -axis was calculated. The

images with good axial symmetry enabled the determination of the electron density to be performed using the Abel inversion. The measurement error was equal to 5%–20%, increasing in the direction from the plasma boundary to the symmetry axis. The total neutron yield was calculated from the data recorded with five silver-activation counters, which were placed at different angles to the discharge z -axis. The first peak of the PIN detector signal was assigned as $t = 0$.

In the next paragraphs, we will evaluate the energy transformations in the plasma column during the phases of its implosion, stagnation, and instability evolution, when the HXRs and neutrons are produced.

A. Energy Balance

The quantity of energy, as delivered from the capacitor bank into the plasma volume, could be evaluated from the recorded voltage, current, and current-derivative waveforms, provided that the inductance is known (see, for example, [8] and [16]). The energy released in the plasma during the period Δt was estimated using the equation

$$UI\Delta t - L_0 I \frac{dI}{dt} \Delta t - L_{CS} I \frac{dI}{dt} \Delta t = \frac{dL_{CS}}{dt} I^2 \Delta t + RI^2 \Delta t. \quad (1)$$

In this equation, the first term on the left side describes the energy input determined from the voltage U and current I signals. The second term describes the energy value of a magnetic field inside the collector and electrode system, in which dI/dt represents the current derivative and the inductance L_0 is supposed to be a constant. The third term describes the magnetic energy of the current sheath of the inductance L_{CS} in front of the electrode system. The second and third terms are positive due to the negative sign of the current derivative at the current drop during the final phase of the current-sheath implosion and elongation. Therefore, the left side of (1) summarizes the energy delivered to the plasma volume in front of the electrodes. The right side describes two forms of energy consumption. The first term describes energy delivered to the plasma due to changes of the inductance of the current sheath (in magnetic and kinetic forms), and the second term describes the energy released in the plasma volume due to its resistance R .

B. Signals of Voltage, Current, and Current Derivative

The signal of voltage had a profile similar to the absolute value of the current derivative, as described in [16]. Usually, during each HXR and neutron pulse, the voltage waveform has a peak, and the current derivative shows a dip. The peaks of voltage and the dip of current derivative are not prominent in all shots. They are dominant in the first pulse and sometimes prominent in the later pulses. The temporal coincidence of their extremes is of order 10 ns, worse than the uncertainty in the temporal correlation of signals (2 ns). Then, the energy delivered to plasma increases immediately before and during the time of fast particle acceleration. As an example, we choose shot #8584. The recorded signals of the current, the current derivative, the HXRs, and the neutrons are shown in Fig. 1.

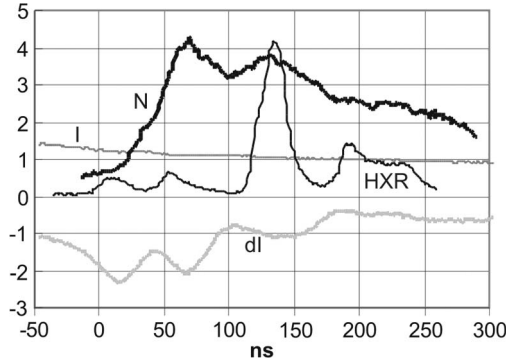


Fig. 1. Shot #8584. The scale of the current trace (I) is in megaamperes, and the current derivative trace (dI) is in teraamperes per second. The signal of HXR and neutrons (N) are plotted in arbitrary units.

In the waveforms of the HXR and current derivative, one could easily identify first three explicit pulses in the periods of 20–30 ns, 40–70 ns, and 120–150 ns, respectively. The neutron signal, which was registered at a distance of 7 m side-on, was recalculated for the instant $t = 0$ by a time-of-flight method, using the velocity 2.16×10^7 m/s of 2.45-MeV energy of neutrons. The neutron peaks are smoothed as a consequence of the energy distribution and scattered neutrons. The neutrons are usually delayed in comparison with HXR regarding the time of flight between their generation and neutron production up to 10–20 ns. The first neutron pulse in this shot was very weak, and its neutron peak was insignificant. As a result, we can see that the first three extremes in Fig. 1 correlate with an accuracy of 10–20 ns.

C. Evaluation of the Inductance and Its Derivative, and Resistance

The form of energy delivered to the plasma volume, i.e., the right side of (1), could be calculated on the basis of the estimated values of the inductance and resistance. The inductance L_0 in the PF-1000 facility, as calculated from the formula [17] for a long coaxial conductor with a cathode composed from tubs, was 32 ± 1 nH. The inductance of the current sheath in front of the electrodes was calculated from a profile of the dense current sheath (recorded on the interferometric images), under the assumption that the skin layer is about 1 cm thin and 80% of the total current flows upon the surface of the dense column and the inner surface of the umbrella-shaped current sheath. The value of inductance varied temporally (during the HXR and neutron production) and from shot to shot in the range of 20–35 nH. The uncertainty of this value was about 10%, taking into account uncertainty in the external diameter of the current sheath, since the interferometer viewed a region smaller than the cathode diameter. The temporal dependence of the current-sheath inductance for shot #8584 is shown in Fig. 2. One can easily see that it reached the maximum during the second pulse, when the dominant portion of fusion neutrons was produced, and later decreases. For a comparison, in Fig. 2, the signals of soft X-rays (SXR) and HXR are also shown.

From the known inductance, voltage, and current values, we could calculate the energy W delivered to the plasma volume

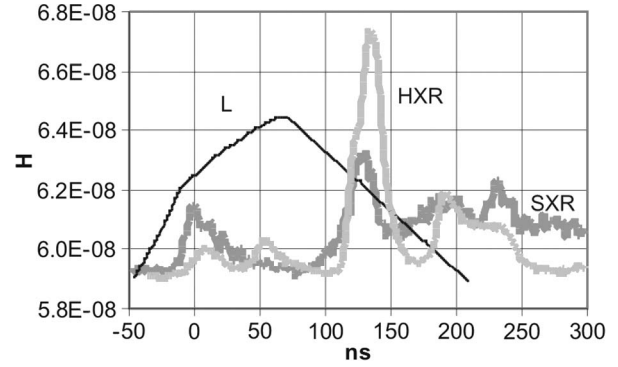


Fig. 2. Shot 8584. Temporal dependence of the current-sheath inductance (L) in comparison with signals of SXR filtered with $10 \mu\text{m}$ of Be foil (SXR) and HXR.

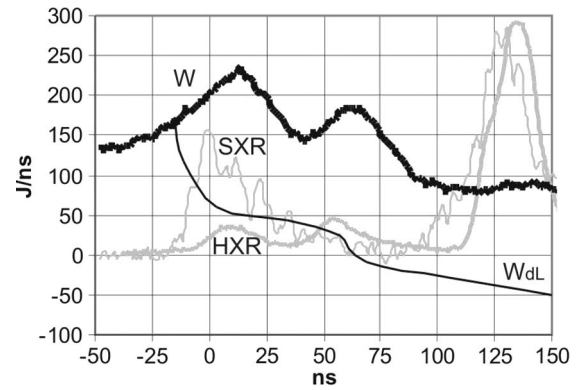


Fig. 3. Shot 8584. Temporal dependence of (thick black) energy W delivered to the plasma column during 1 ns and (thin black) energy of the inductance derivative W_{dL} in comparison with (thin shadow) SXR and (thick shadow) HXR.

in front of the electrode, which corresponded to the sum of energies on the left side of (1). The temporal dependence of this energy per 1 ns, as calculated for shot #8584, is shown in Fig. 3. A 30% uncertainty of the W value was caused mainly by the uncertainty of inductance calculations.

The considered energy can be transformed, as mentioned earlier, through two channels, which are described by both of the terms on the right side of (1).

The first term describes an increase in magnetic energy in the plasma volume during an increase in the inductance (caused by the compression of the plasma column and an increase in its length) and, after that, an energy decrease during the decrease in the inductance (caused by the plasma column expansion). The derivative of the inductance can be calculated from the temporal changes of the inductance. The interferometric images were recorded with temporal intervals of 10–20 ns. Hence, the uncertainty of the inductance derivative was increased (due to the uncertainty of the interpolation) to about 20%. The temporal dependence of energy W_{dL} , expressed by the first term on the right side of (1) and computed per 1 ns for the shot #8584, is shown in Fig. 3. The uncertainty of the energy W_{dL} value is about 30%.

It has already been mentioned that the second term in (1) describes the energy transfer through the resistance to the charged particles carrying the discharge current. The R_0 of the

circuit was 2.6 m Ω . The Spitzer resistance of the plasma was estimated below $2.8 \times 10^4 \Omega^{-1} \text{ m}^{-1}$, from the known length of the current sheath (equal to about 20 cm), the skin layer thickness is equal to about 1 cm (measured with magnetic probes), and the conductance for the electron temperature is above 50 eV (as the lower limit), under the assumption of a constant current density in the current sheath. In this case, the resistance of the plasma volume in front of the electrodes should be below $2 \times 10^{-2} \Omega$, and the energy deposited by the Joule heating for the current of 1 MA should be below 20 J per 1 ns. The same value is achieved by the Bremsstrahlung from plasma with a temperature in the range of 50–300 eV and the mean density of about $4 \times 10^{24} \text{ m}^{-3}$. Hence, the Joule heating corresponding to the Spitzer resistance could amount to 5%–10% of the total energy only.

Therefore, the difference between the W and W_{dL} values in the range of 100–200 J/ns should be delivered with an anomalous resistance probably due to turbulence [18]. This energy was delivered just before and during the acceleration of fast particles and varied shot to shot in the range of 3–10 kJ. The uncertainty of the difference between W and W_{dL} was estimated to be 60%.

D. Final Phase of Implosion of the Current Sheath and the Initial Phase of Stagnation

In the described PF-1000 experiments, the current sheath in front of the electrodes imploded during 500–600 ns. In the final phase of the implosion, we could estimate the velocity and number of particles from the recorded interferometric images. For the linear density equal to $(1\text{--}2) \times 10^{19}$ ions/cm and the plasma velocity of $(2.2\text{--}2.5) \times 10^5$ m/s, the final kinetic energy value reached 1–1.5 kJ/cm. This kinetic energy heated the plasma. Therefore, the temperature of ions at the start of the stagnation process could be about 600 eV. The relaxation time of such ions, as evaluated from the Spitzer formulas [19], was about 10 ns, while the duration of heating of electrons by the fast deuterons was longer and amounted to about 100 ns. Hence, the temperature of electrons was lower than that of ions, and its upper limit was evaluated to be 200–300 eV.

The energy of the magnetic field of the axial current going through the surface at the stagnation was about 1 kJ/cm.

The axial position of the minimal diameter of the plasma column was affected by zippering, which moves from the anode end to a distance of 6 cm in the period lasting from -8 to 100 ns, with the mean velocity of about 5×10^5 m/s. During this time, the portion of the plasma column near the anode expanded in the radial directions with a velocity of $(1.5\text{--}2.0) \times 10^5$ m/s, while the other portion (near the umbrella-shaped current sheath) continued in the implosion. Both motions influenced the resulting inductance and energy balance in the plasma column.

E. First Neutron and HXR Pulse

The first neutron and HXR pulse, as observed in the initial phase of the stagnation process, started usually at ~ -10 ns together with the SXR at the instant of the minimal diameter

of the pinch column near the anode end, and it ended about 30–40 ns later. As an example, the results from the shot #8584 are shown in Figs. 1–3. The total neutron yield in the considered weak and lengthened pulse amounted to about 3×10^{10} , and the maximal neutron energy (measured downstream) reached about 2.6–2.7 MeV. Knowing the density and dimensions of the plasma region and the mean energy of fast deuterons responsible for the production of the fusion neutrons [15], one could estimate that the number of the fast partially circulating deuterons (more exactly its lower limit) was equal to 10^{17} , and the total energy of ion beams was about 1–2 kJ. The uncertainty of this energy value of factor 2 was caused mainly by an uncertainty in the estimations of the number of deuterons, having energies in the range of up to a few tens of kiloelectronvolts and a low DD cross section. In any event, the energy delivered during the first pulse (through the anomalous resistance) was about 4–5 kJ. Hence, in the column, there was enough energy for the acceleration of the fast particles (deuterons) producing the first neutron pulse.

F. Second Neutron and HXR Pulse

The second neutron and HXR pulse was observed during the stagnation and the increase in the pinch diameter. As an example, we considered the shot #8584 once again shown in Figs. 1–3, which produced the total neutron yield equal to 10^{11} and which was described in more detail in [15]. In that paper, we presented estimations of the total energy of fast deuterons during the second neutron pulse. The energy of the fast deuterons responsible for the production of the dominant number of fusion neutrons was evaluated similarly as in Section II-E, i.e., 3 kJ. Similarly, an error about 100% was caused mainly by the uncertainty in the determination of the energy distribution of the fast deuterons. In this instance, the energy contained in the fast deuterons was considerably higher than the energy contained in the pinch (in the form of kinetic energy of deuterons and in the magnetic field). The energy delivered to the plasma during 30–80 ns (see Fig. 3) was higher than 10 kJ and had to be delivered during a few tens of nanoseconds, probably through the anomalous resistance. Similar results were obtained during the analysis of intense second pulses in other PF-1000 shots.

G. Neutron and HXR Pulse at the Constriction

This process of the HXR and neutron production in the shot #8596 was analyzed in the previous paper [15] as well. In that case, the HXRs and neutrons were produced during the final phase of the stagnation at $t = 100\text{--}120$ ns. The total neutron yield was 10^{11} , and the energy of the fast deuterons producing neutrons was estimated to be 3 kJ (with an error of 100%).

For the shot #8596 (shown in Fig. 4), we could also evaluate the energy released at the implosion of the pinch column neck. In the case considered, the axial length of the imploding plasma column was equal to about 1.5 cm. The mean velocity of the imploded deuterons was 1.8×10^5 m/s, and their mean kinetic energy amounted to about 300 eV. The total number of particles in the pinch constriction before its explosion was estimated to

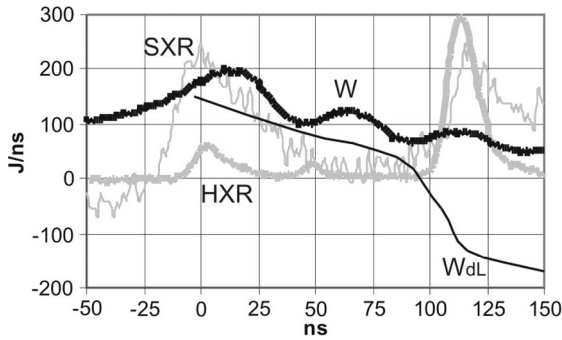


Fig. 4. Shot #8596. Temporal dependence of energy W computed per 1 ns and energy connected with the inductance derivative W_{dL} in comparison with the SXR and HXR.

be 1.5×10^{19} . As a result, the total kinetic energy of ions in the constriction could amount to only 0.5 kJ, a figure insufficient for the fast deuteron acceleration.

The total energy delivered to the plasma could be evaluated from Fig. 4 as the difference between the W and W_{dL} values. During the period of 40–120 ns, that difference amounted to 6 kJ with an error of 60%. Consequently, one might conclude that the energy carried out by the fast electrons and ions could be delivered to the plasma mainly through the anomalous resistance immediately before and during their acceleration.

III. DISCUSSION AND CONCLUSION

The presented conclusions about the dominant role of anomalous resistance have been obtained on the assumption of a current flowing through the dense column surface. From interferometry diagnostics, the existence of toroidal, helical, and spherical plasmoidal structures follows, in which self-generated closed currents and magnetic fields can be present. These phenomena can cause a complicated direction of the current and increase the inductance. This scenario should be more probable during the first pulse, when the imploded current sheaths, particularly if they have a filamentary structure, can penetrate into the inner layers, increasing the internal energy. Then, at the assumption of constant current density, the energy should increase about 100–200 J/ns. During the later pulses, the assumption of the surface current seems to be more probable, and the anomalous resistance seems to be more reasonable. Moreover, in cases of later outstanding extremes of voltage and current derivative, we observe, in interferograms of imploding constrictions and formatting of a plasmoids, broken fringe pattern as the possible consequence of turbulence.

The conclusion can be summarized as follows.

The time-resolved laser interferometry and scintillation detectors, used together with routine measurements of the voltage, current, and current-derivative waveforms, enabled us to evaluate the energy delivered and released in the dense plasma column during the period of the HXR and neutron production, when the energetic beams of deuterons and electrons are also produced.

The inductance of the current sheath in front of the electrode outlets was evaluated from a profile of the dense current sheath, which was observed on the recorded interferometric images,

under the assumption of a thin skin layer. It was observed that the peaks of the delivered power corresponded to the period of the deuteron and electron acceleration. The energy delivered to the plasma volume during one power peak was estimated to be about 3–10 kJ.

The fast deuterons producing intense fusion neutron pulses had energy in the similar range. This energy in intensive pulses was higher than the energy stored in the pinch in the form of kinetic and magnetic energies. As a result, it was deduced that the significant portion of energy necessary for the charged particle acceleration, which was estimated to be 3–5 kJ, should be delivered to the pinched plasma in a few dozen nanoseconds before and during the HXR and neutron pulse, mainly through anomalous resistance.

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Pavel Kubes, photograph and biography not available at the time of publication.

Daniel Klir, photograph and biography not available at the time of publication.

Jozef Kravarik, photograph and biography not available at the time of publication.

Karel Rezac, photograph and biography not available at the time of publication.

Marian Paduch, photograph and biography not available at the time of publication.

Tadeusz Pisarczyk, photograph and biography not available at the time of publication.

Marek Scholz, photograph and biography not available at the time of publication.

Tomasz Chodukowski, photograph and biography not available at the time of publication.

Barbara Bienkowska, photograph and biography not available at the time of publication.

Irena Ivanova-Stanik, photograph and biography not available at the time of publication.

Lesław Karpinski, photograph and biography not available at the time of publication.

Marek Jan Sadowski, photograph and biography not available at the time of publication.

Krzysztof Tomaszewski, photograph and biography not available at the time of publication.

Ewa Zielinska, photograph and biography not available at the time of publication.